

CSE 485 Final Exam

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I. QUESTION

Define a data mining application and show briefly the appropriate steps (preprocessing, association rule mining, classification, prediction etc.) for small example data.

II. QUESTION

Suppose that the data for analysis include the attribute age. The age values for data tuples are: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- Use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.
- Use normalization by decimal scaling to transform the value 35 for age.
- Sketch examples for each of the following sampling techniques: SRSWOR, SRSWR. Use samples of size 5.

III. QUESTION

The Apriori algorithm makes use of prior knowledge of subset support properties.

- Prove that all nonempty subsets of a frequent itemset must also be frequent.
- Prove that the support of any nonempty subset s' of itemset s must be as great as the support of s .
- A partitioning variation of Apriori subdivides the transactions of a database D into n nonoverlapping partitions. Prove that any itemset that is frequent in D must be frequent in at least one partition D .

IV. QUESTION

The following table consists of training data from an employee database. Let status be the class label attribute.

Use ID3 to construct a decision tree from the given data.

Department	Status	Age	Salary
sales	senior	31..35	46K..50K
sales	junior	26..30	26K..30K
sales	junior	31..35	31K..35K
systems	junior	21..25	46K..50K
systems	senior	31..35	66K..70K
systems	junior	26..30	46K..50K
systems	senior	41..45	66K..70K
marketing	senior	36..40	46K..50K
marketing	junior	31..35	41K..45K
secretary	senior	46..50	36K..40K
secretary	junior	26..30	26K..30K

V. QUESTION

The following table shows the midterm and final exam grades obtained for students in a database course.

Midterm Exam	Final Exam
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
88	74
81	90

- Use the method of least squares to find an equation for the prediction of a student's final exam grade based on the student's midterm grade in the course.
- Predict the final exam grade of a student who received an 86 on the midterm exam.

VI. QUESTION

Suppose that the data mining task is to cluster the following eight points (with (x, y) representing location) into three clusters. $A_1(2, 10)$, $A_2(2, 5)$, $A_3(8, 4)$, $A_4(5, 8)$, $A_5(7, 5)$, $A_6(6, 4)$, $A_7(1, 2)$, $A_8(4, 9)$. The distance function is Euclidian distance. Suppose initially we assign A_1 , A_4 , and A_7 as the center of each cluster, respectively. Run the k-means algorithm on this data.